

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |                              |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------|
| 7a | <p>When the <u>rate of decay of Xe-143 is faster than that of Cs-143</u>, the number of Cs-143 <u>increases</u>.</p> <p>When the <u>rate of decay of Xe-143 decreases, as number of Xe-143 decreases</u>, till it is <u>below the rate of decay of Cs-143</u>, the <u>number of Cs-143 decreases</u>.</p> <p>Or</p> <p>When the <u>rate of decay of Cs increases, as number of Cs increases</u>, till it is <u>above the rate of decay of Xe</u>, the <u>number of Cs-143 decreases</u></p> | A | B1<br><br><br>A1<br><br>(A1) |
| 7b | Due to the <u>spontaneous</u> nature of radioactivity decay, the rate of decay is unaffected by external environment such as surrounding temperature.                                                                                                                                                                                                                                                                                                                                       | A | A1                           |
| 7c | <p>The number of radiations does not just come from Xenon, but also from Caesium.</p> <p>The count rate may include background radiation.</p>                                                                                                                                                                                                                                                                                                                                               | A | B1<br><br>B1                 |
| 7d | $\beta$ -particles have high speeds/energy                                                                                                                                                                                                                                                                                                                                                                                                                                                  | D | B1                           |

|  |                                                                               |  |    |
|--|-------------------------------------------------------------------------------|--|----|
|  | when charged particle is accelerated/slowed down                              |  | M1 |
|  | electromagnetic radiation is produced                                         |  | A1 |
|  | $\beta$ -particles stopped in the lead produce X-ray radiation which escapes. |  | B1 |

|     |                                             |   |    |
|-----|---------------------------------------------|---|----|
| 8ai | Heat lost by water = heat gained by mercury | A | C1 |
|-----|---------------------------------------------|---|----|